

Human Development Index (HDI) Predictor

Performance Testing

Project: Human Development Index (HDI) Predictor
Phase: Project Testing

1. Model Performance Testing

Metric	Value
R-squared (R^2)	≈ 0.99
Mean Absolute Error (MAE)	≈ 0.013
Root Mean Squared Error (RMSE)	≈ 0.014
Test set size	10% of 144 countries (held out)

2. Why R^2 Was the Primary Metric

HDI prediction is a regression problem with a continuous target between 0 and 1, so R^2 , MAE, and RMSE are the appropriate metrics rather than classification metrics like accuracy or F1-score. An R^2 close to 1 confirms the model explains nearly all of the variance in HDI from the four input indicators.

3. Web Application Testing

Test Case	Input	Expected Result	Actual Result	Status
TC-01	Norway: Life Exp 83.3, Schooling 13.1, GNI 82000, Internet 99	Very High HDI	0.962 — Very High	Pass
TC-02	Niger: Life Exp 61.6, Schooling 2.1, GNI 1300, Internet 10	Low HDI	0.355 — Low	Pass
TC-03	Empty form submission	Validation error, form not submitted	Browser shows required-field error	Pass
TC-04	Home page GET request	Renders introduction page	HTTP 200, home.html rendered	Pass
TC-05	Flask server startup	Server starts on port 5000	Running on http://127.0.0.1:5000	Pass

4. Data Pipeline Testing

Test	Expected	Result	Status
generate_dataset.py runs without error	HDI.csv created, 144 rows	144 rows written	Pass
Null value handling	0 missing values in X after fillna	0 total missing values	Pass
Country label encoding	All 144 countries encoded uniquely	144 unique classes	Pass
HDI.pkl saved correctly	File exists and loads without error	Loaded successfully	Pass

5. Performance Benchmarks

Benchmark	Result
Model training time	Under 5 seconds on a standard laptop
Prediction response time	Under 0.3 seconds per request
Flask startup time	Under 2 seconds, including model load

Benchmark	Result
Dataset size	144 countries, 9 columns